Marriott International

Hotel Revenue Analytics

How can reservation data be structured and exploited to analyse and improve hotel profitability?

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Marriott International – Company Overview

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brands

Hotel Brands

THE RITZ-CARLTON → COURTYARD Marriott

9,000+

hotels

Properties Worldwide

$22.73B

revenue

Annual Revenue (2025)

150+

countries

Countries & Territories

Why Marriott?

Marriott manages massive reservation data across its properties.
My interest in luxury hotels led me to focus on this group.
The real challenge is not collecting data, but using it to drive profitability.

Financial Problem & Database Choice

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The Financial Problem

Core question:

<How can reservation data be structured & exploited to analyse and improve hotel profitability?>

Key financial KPIs:

- RevPAR
- ADR (Average Daily Rate)
- Occupancy Rate
- Revenue per booking channel
- Seasonal revenue trends

DBMS Choice: PostgreSQL

Why a Relational DBMS?

→ **Structured data:** reservations, rooms, hotels, revenues all have fixed schemas.

→ **SQL aggregations:** ideal for GROUP BY, SUM, AVG on financial metrics.

→ **Referential integrity:** foreign keys prevent orphan records.

Why PostgreSQL specifically?

Free, production-grade, window functions for RevPAR calculations, Python connector (psycopg2).

Database Schema — Entity Relationship Overview

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Hotels

```
hotel_id (PK)
name
city
country
category
```

→ FK

Rooms

```
room_id (PK)
hotel_id (FK)
room_type
base_price
capacity
```

→ FK

Customers

```
customer_id (PK)
name
loyalty_tier
country
```

↓ FK

Reservations

```
reservation_id (PK)
room_id (FK)
customer_id (FK)
check_in
check_out
channel
total_revenue
```

→ FK

↓ FK

Revenue_Facts

```
fact_id (PK)
reservation_id (FK)
hotel_id (FK)
date
RevPAR
ADR
occupancy_rate
```

PK = Primary Key | FK = Foreign Key | All KPIs computed via SQL aggregations

Exploiting the Data – SQL Queries & Dataset

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Key SQL Queries

```
-- RevPAR by hotel & month
SELECT h.name,
      DATE_TRUNC('month', r.check_in) AS month,
      SUM(r.total_revenue) /
        COUNT(DISTINCT ro.room_id) AS RevPAR
FROM reservations r
JOIN rooms ro ON ro.room_id = r.room_id
JOIN hotels h ON h.hotel_id = ro.hotel_id
WHERE r.is_cancelled = FALSE
GROUP BY 1, 2
ORDER BY 2, RevPAR DESC;

-- Revenue by booking channel
SELECT channel,
      COUNT(*)           AS bookings,
      SUM(total_revenue) AS total_rev,
      AVG(total_revenue) AS avg_rev
FROM reservations
WHERE is_cancelled = FALSE
GROUP BY channel
ORDER BY total_rev DESC;
```

Dataset Used

Source: Kaggle – Hotel Booking Demand

119,390 records

Columns: room type, lead time, channel, ADR, cancellations…

Example Insights Unlocked

Direct bookings → +18% ADR

July & August peak RevPAR

Loyalty tiers cut cancellations

Suite rooms under-priced vs demand

Conclusion & Key Takeaways

1. Marriott needs a relational DBMS to link reservations, rooms, hotels and financial KPIs.
2. PostgreSQL is the ideal choice: free, powerful SQL aggregations, Python-compatible.
3. My schema enables direct computation of RevPAR, ADR and occupancy rate from raw data.
4. Booking channel analysis shows direct bookings are 18% more profitable than OTAs.
5. A well-designed DBMS is not just storage — it is a financial decision-support tool.

Thank you — Questions?